CIS020-1 - Introduction to Software Development
CIS093-1 – Mathematics and Concepts for Computational Thinking

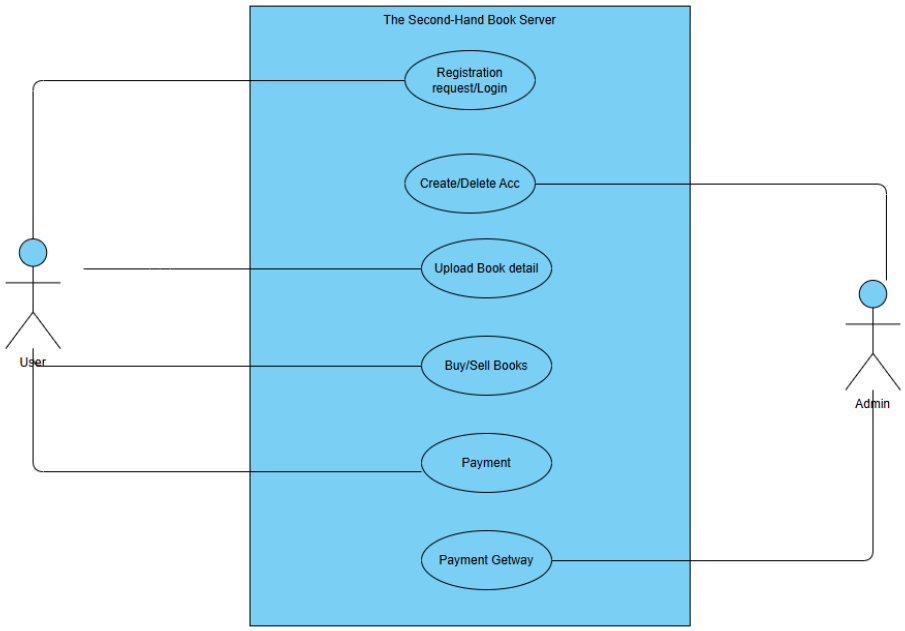
UML Workshop 2 - Adding Relationships Use Case diagrams

Part 1

Consider how you would create a Use Case diagram for the following problem. Then create a diagram with markings to indicate the appropriate relationships. Add notes to your diagram to clarify your view and state any assumptions you make.

The Second-Hand Book Server

This system allows people with books they would like to sell to contact people who might be interested in buying them. Everybody who uses the system needs to have an account. Only the administrator can create or delete the accounts. Once a user has an account they can, if they wish, post a book for sale. They can search the list of books currently available. If they see a book they like they may purchase it which causes the list to be updated.

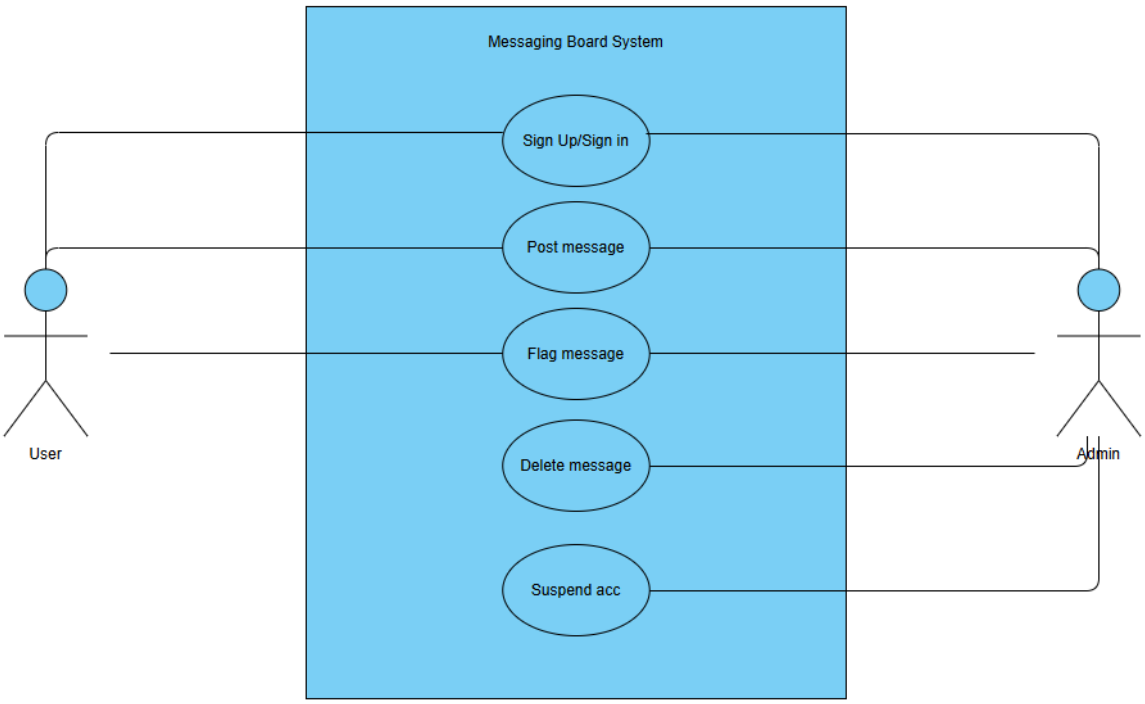


Part 2

Consider how you would create a Use Case diagram for the following problem. Then create a diagram with markings to indicate the appropriate relationships. Add notes to your diagram to clarify your view and state any assumptions you make.

Messaging Board System

This system users allows to post messages interactively to a common online forum. The systems is community-run so the administrators are also users of the system. While the users can post messages, the administrators can also delete them. If a user flags a post as being abusive it is referred to the administrators for vetting and, if it is found to be abusive, the user may have their membership suspended. Users must log on before using the system and if they are suspended they will be turned away at this stage.



Part 3

Consider an ATM system. Who are the actors and what are the Use Cases which describe its behaviour? Represent these on a Use Case diagram. You are encouraged to use the example of an ATM you have used rather than any formed examples. Now consider how you would model the following relationships in your use case diagram using Include, Extend and Generalisation structures.

The ATM machine requires a pin code before it provides access to services.

The services provided (Check balance, Withdraw funds, Print statement) have a common profile and must manage possible failure modes.

Customers who belong to the ATM's bank are additionally able to transfer funds from the machine.

